



Implementations of Quantum and Classical Topology-Aligned Architectures for Molecular Property Prediction

IEEE Quantum Week Toronto

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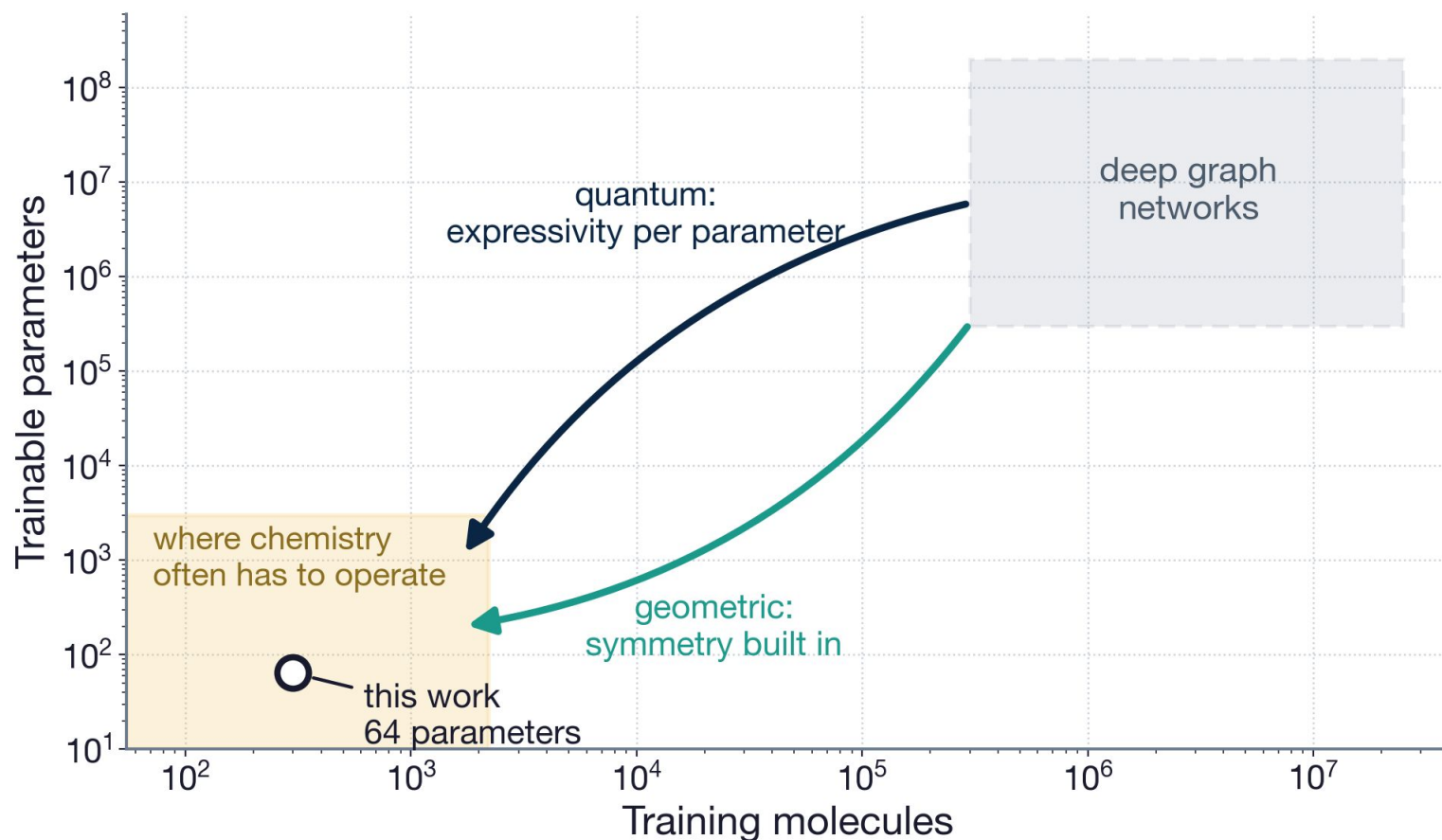
Introduction

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Data & Parameters.

The exploration of chemical space has become a data and parameters problem

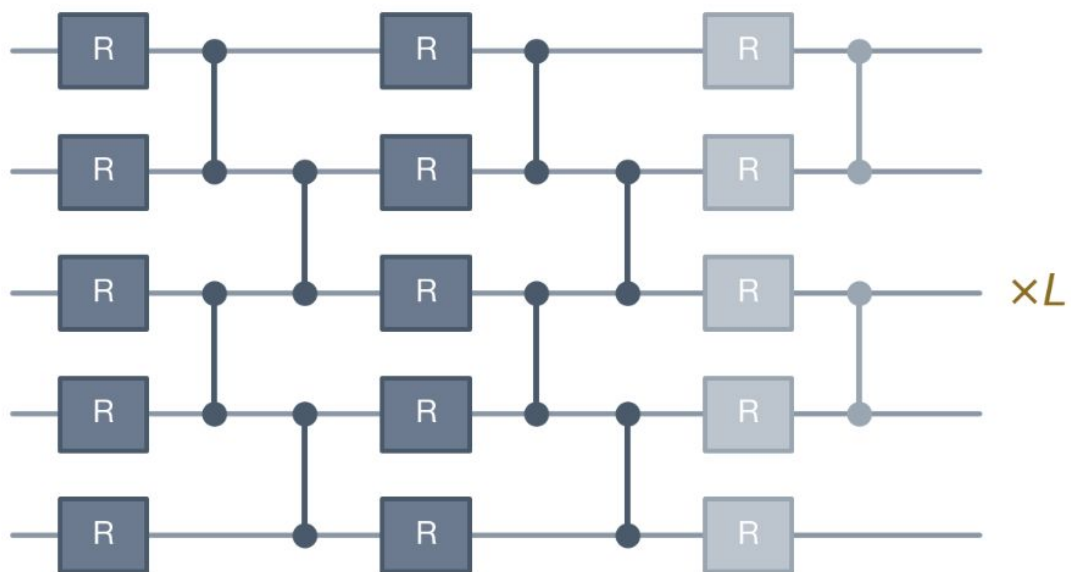
- **Intricate electronic structure:** cost explodes
- **Deep networks:** more parameters, more data
- **Quantum:** expressivity per parameter
- **Geometric:** symmetry built in



Introduction

Contributions.

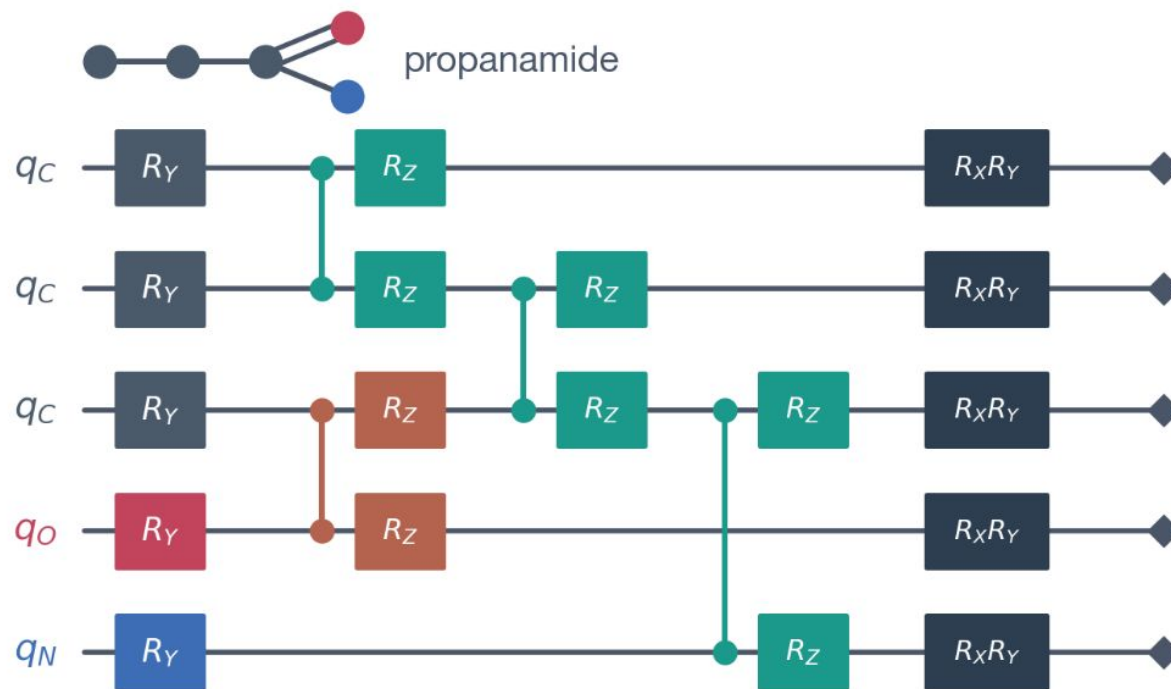
Hardware-efficient ansatz



gates follow the processor

depth must grow to capture chemistry

Topology-aligned (this work)



gates follow the molecule

one layer · 64 parameters

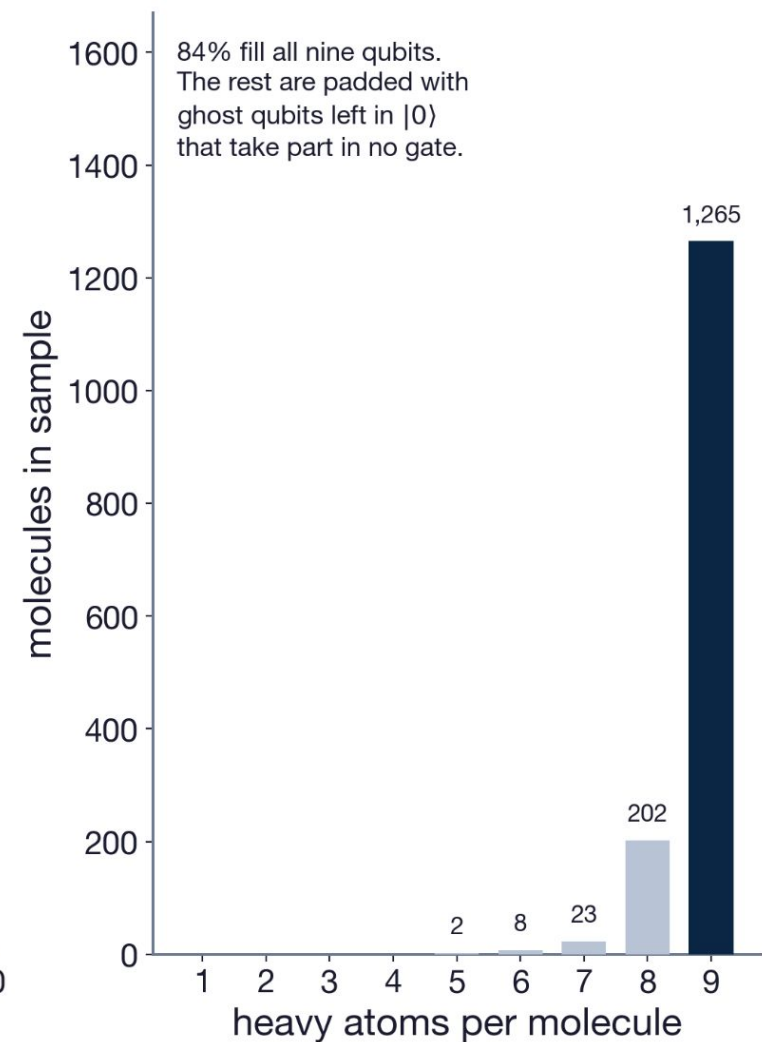
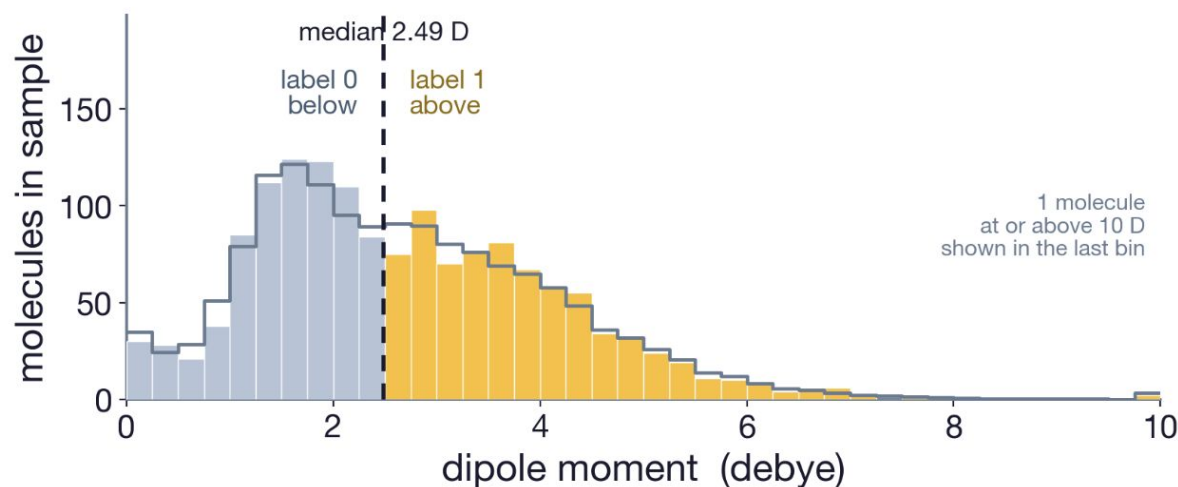
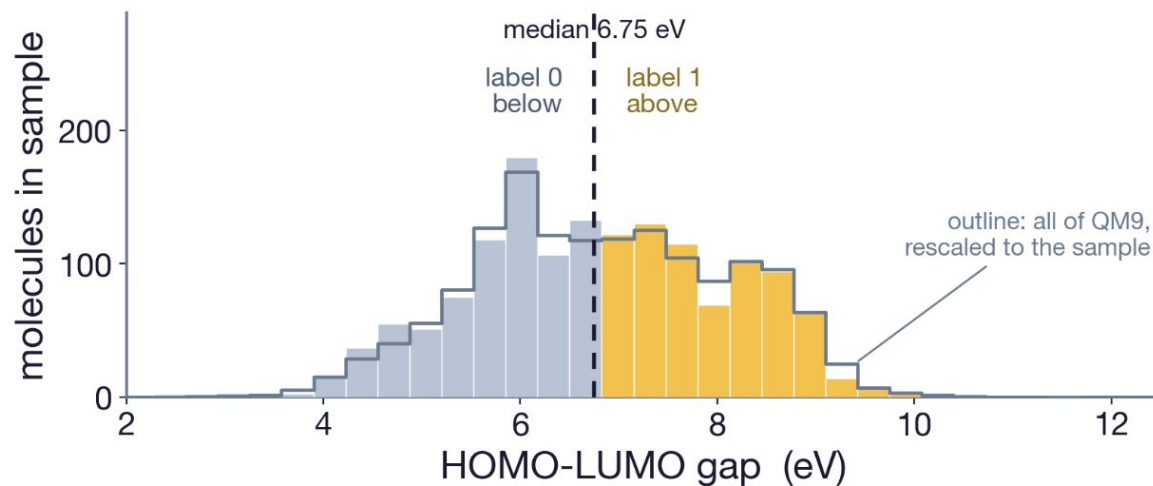
nine-qubit register; the four ghost qubits are not drawn

colour marks a shared parameter: one per atom type, one per bond type

Methodology

Methodology

Dataset and Graph Representation.



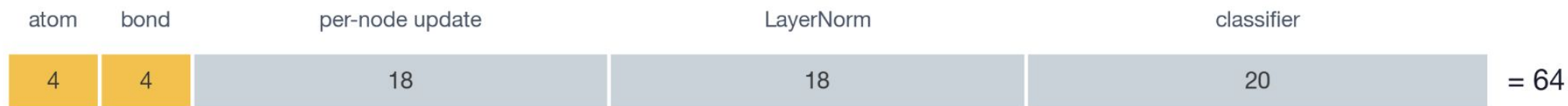
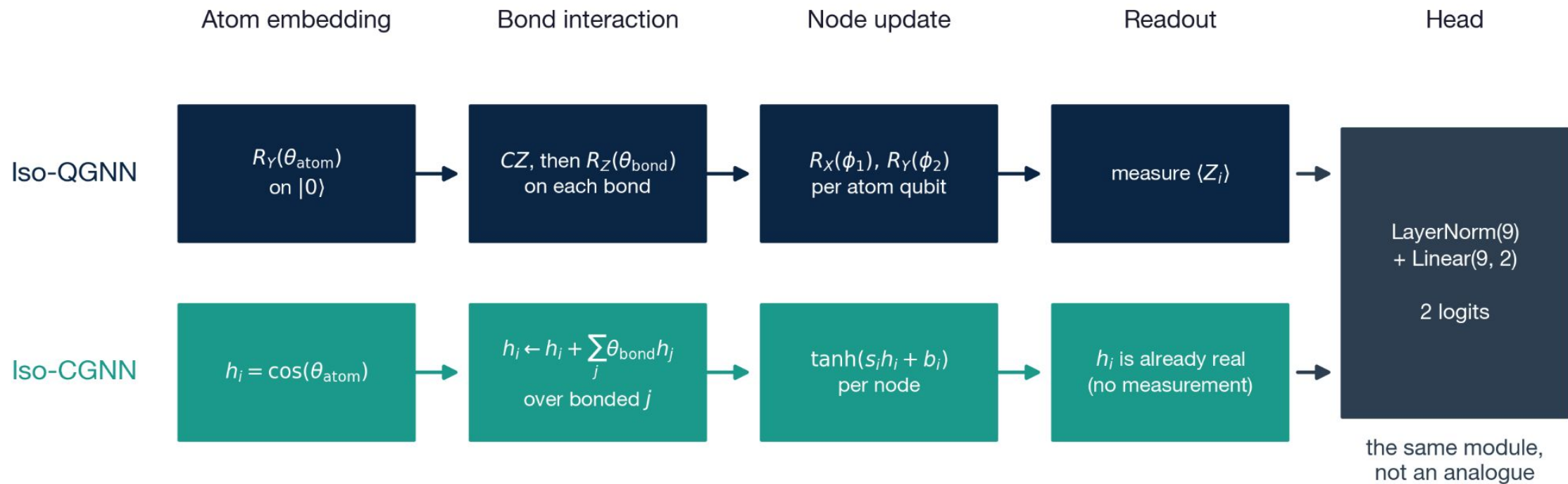
1,500 molecules drawn once from QM9 with a fixed seed, split 1,200 / 150 / 150 in every trial.

Label 1 = above the median of the TRAINING split, recomputed each trial and applied unchanged to held-out molecules.

Dashed lines: whole-sample medians. The per-trial training medians all lie within 0.1 eV and 0.1 D of them.

Methodology

Topology-Aligned Architectures.

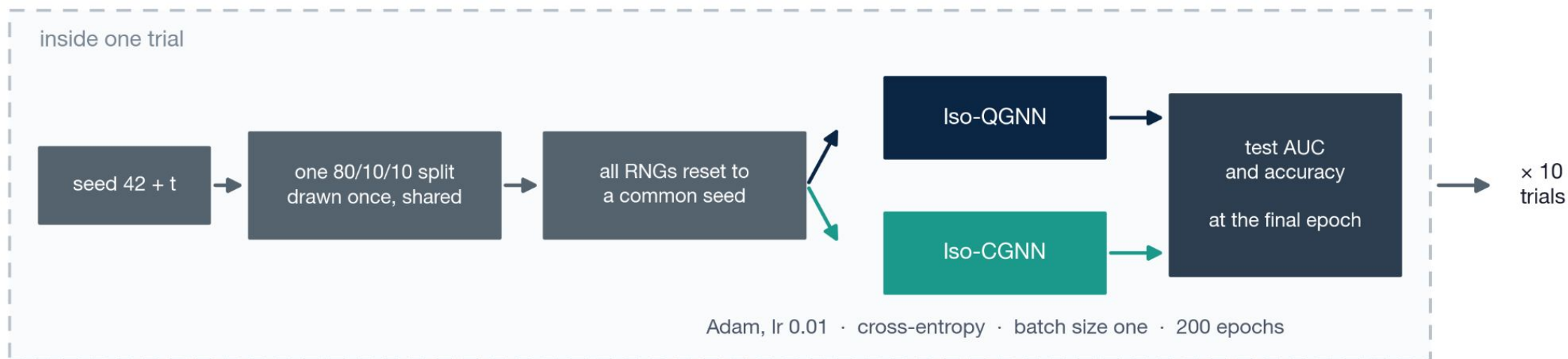


8: the inductive bias

56: shared machinery, identical in design and at initialisation

Methodology

Training Protocol.



Reported: mean and cross-trial standard deviation over the ten trials, seeds 42 to 51. No model selection of any kind.

	matched comparison	sample-complexity sweep
trials per configuration	10	5
epochs	200	40
epoch reported	the final one	chosen on validation AUC
test set	150 held out per trial	a fixed 500 molecules

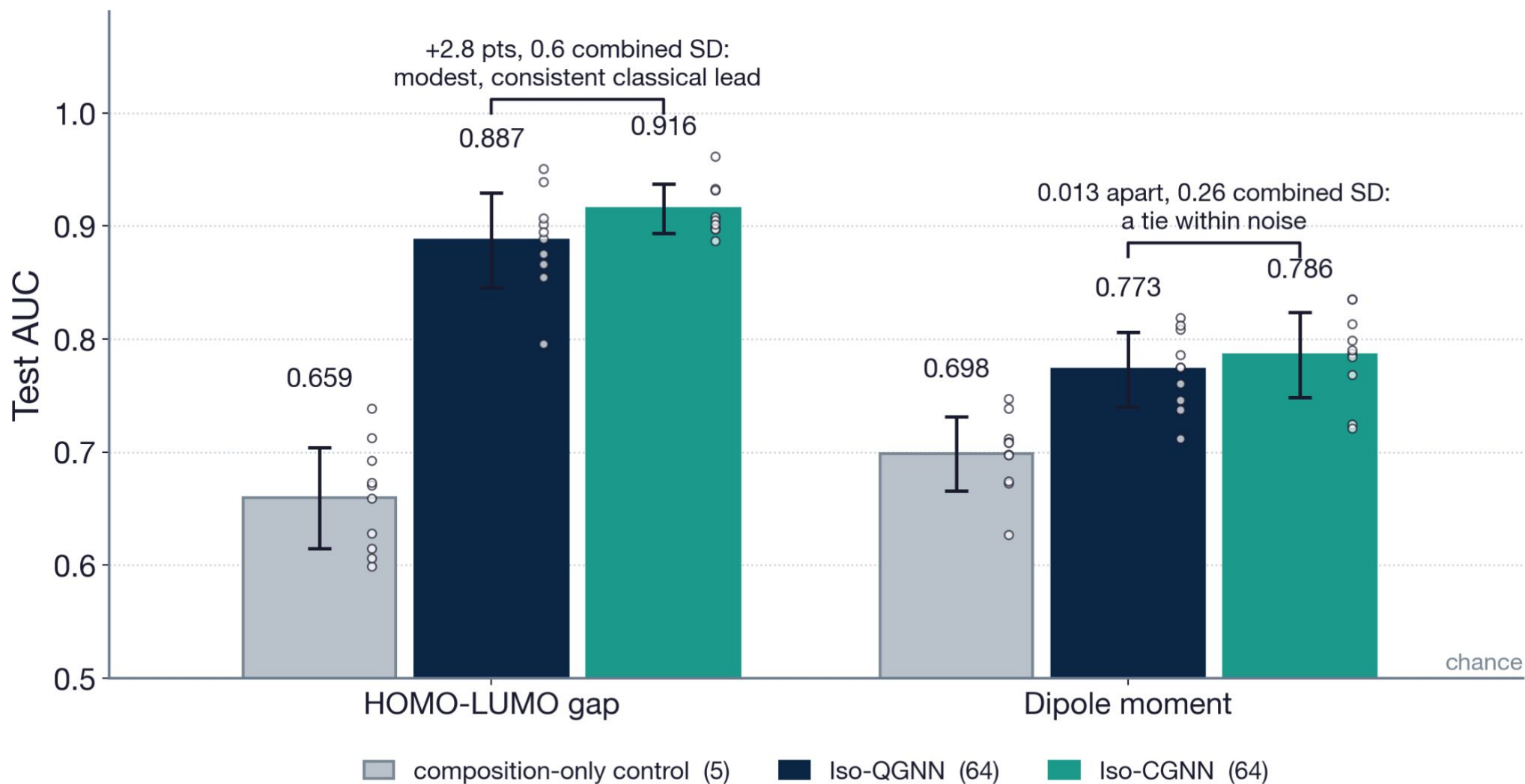
380 trainings against 40, which is why the sweep is shorter. Absolute AUCs are not comparable between the two.

Results



Results

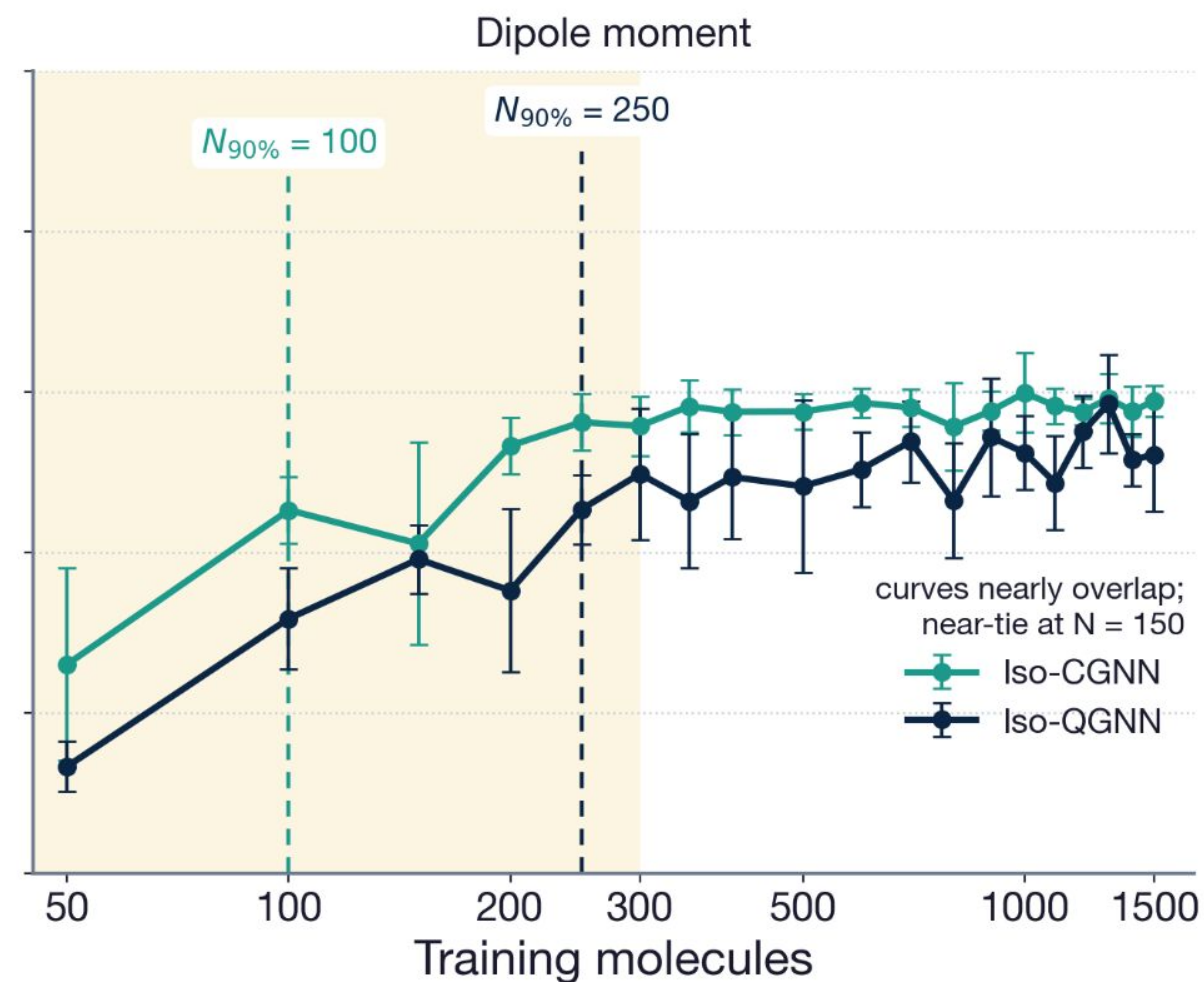
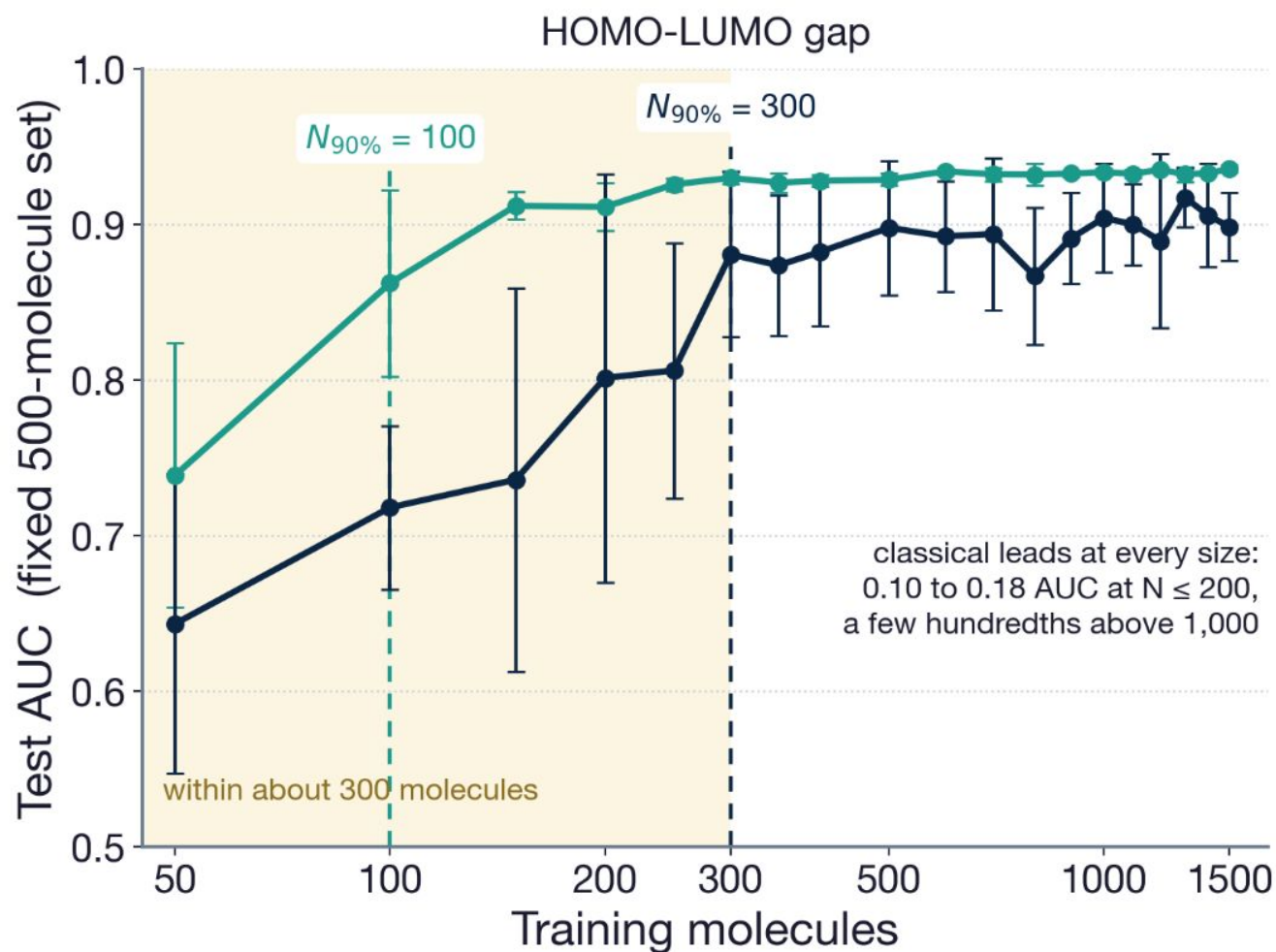
Predictive Performance at Matched Capacity.



Bar: mean over ten trials on identical splits. Whisker: $\pm$ one cross-trial SD.
Dots: the ten trials, one column per bar, height = that trial's AUC.

Results

Sample Complexity.

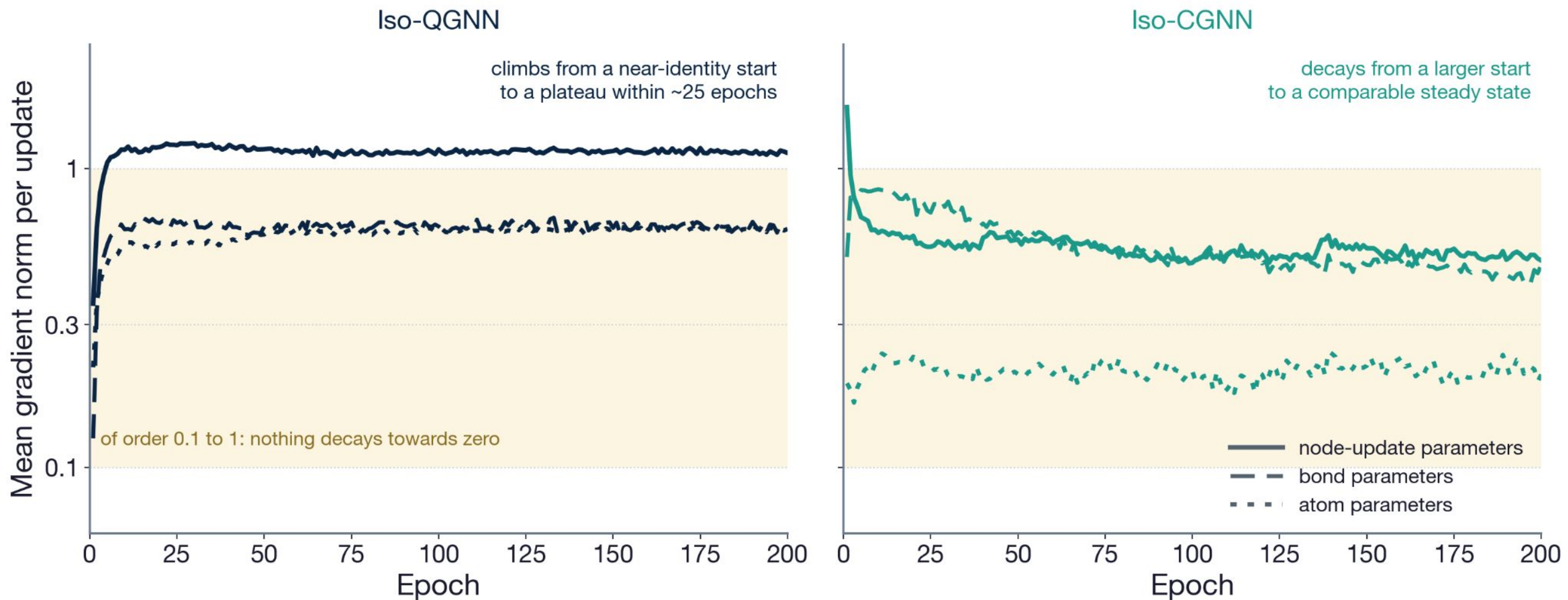


3,000-molecule pool, fixed 500-molecule test set, five trials per size, 40 epochs, epoch chosen on validation AUC.

A different protocol from the matched comparison, so absolute AUCs are not comparable with it. Error bars: $\pm$ one cross-trial SD.

Results

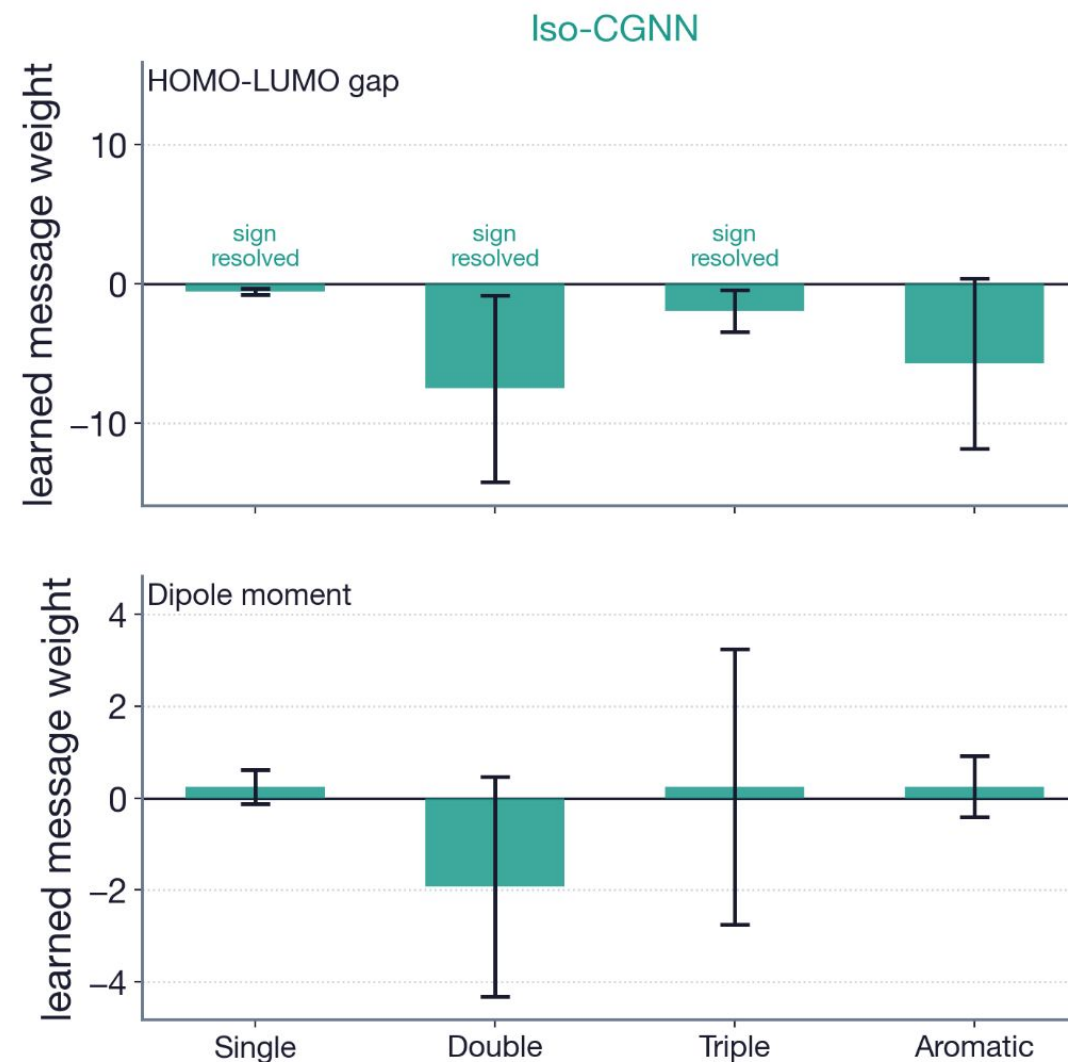
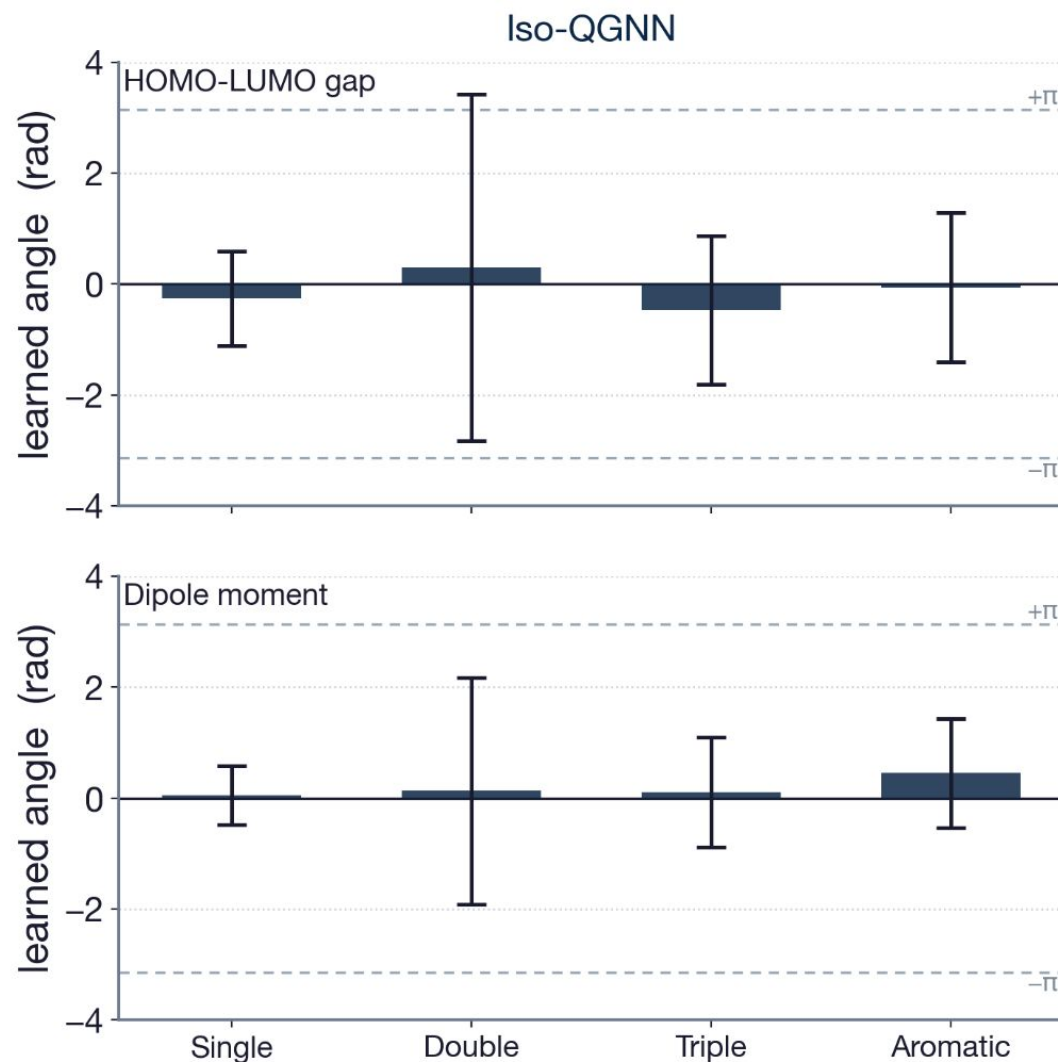
Gradient Stability.



HOMO-LUMO gap task, ten trials. Exact state-vector gradients, no shot noise. Head parameters not tracked.
Nine qubits, one layer: a check at this scale, not a statement about scaling.

Results

Bond-Parameter Analysis.



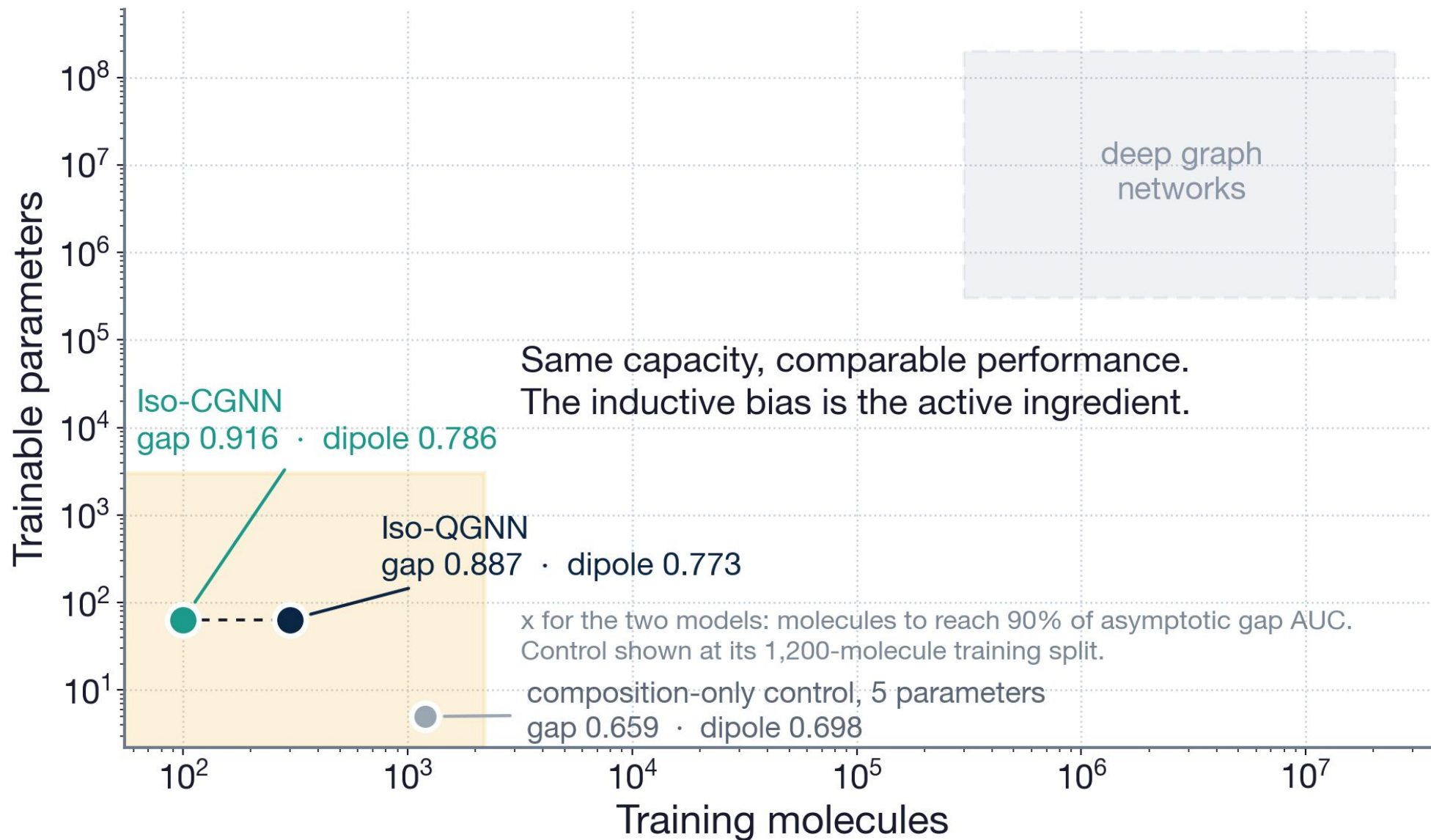
Ten trials, epoch 200. Bars: mean bond parameter. Error bars: $\pm$ one cross-trial SD.
13 of 16 error bars cross zero, so the SD exceeds the mean and even the sign is not determined across seeds.



Conclusions & Future Work

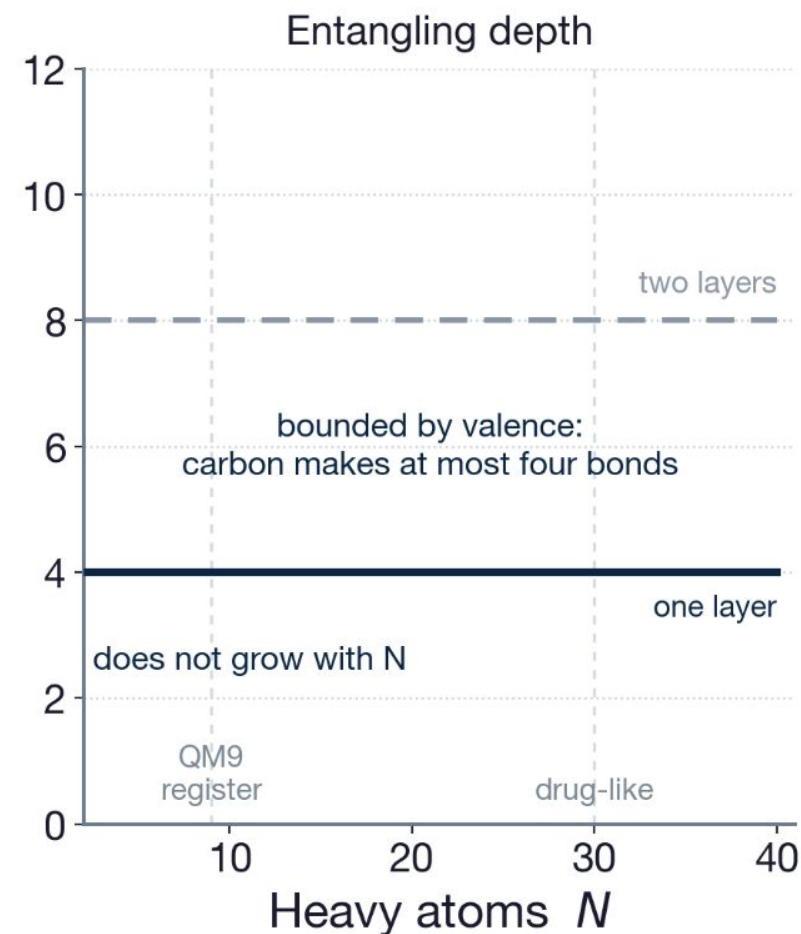
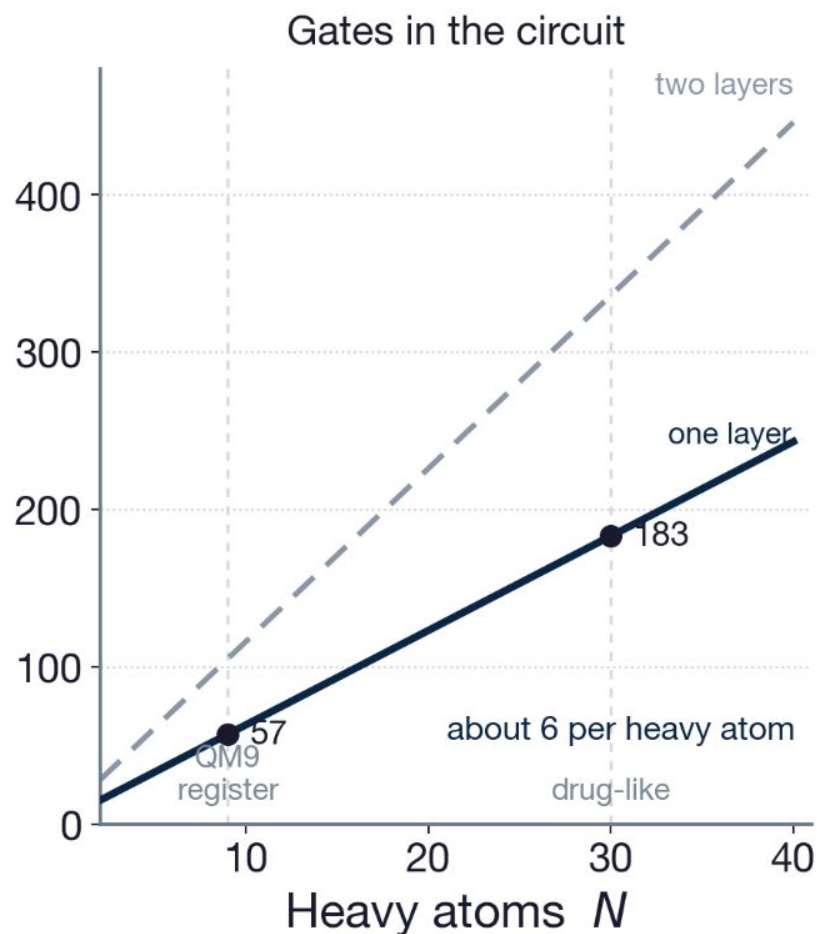
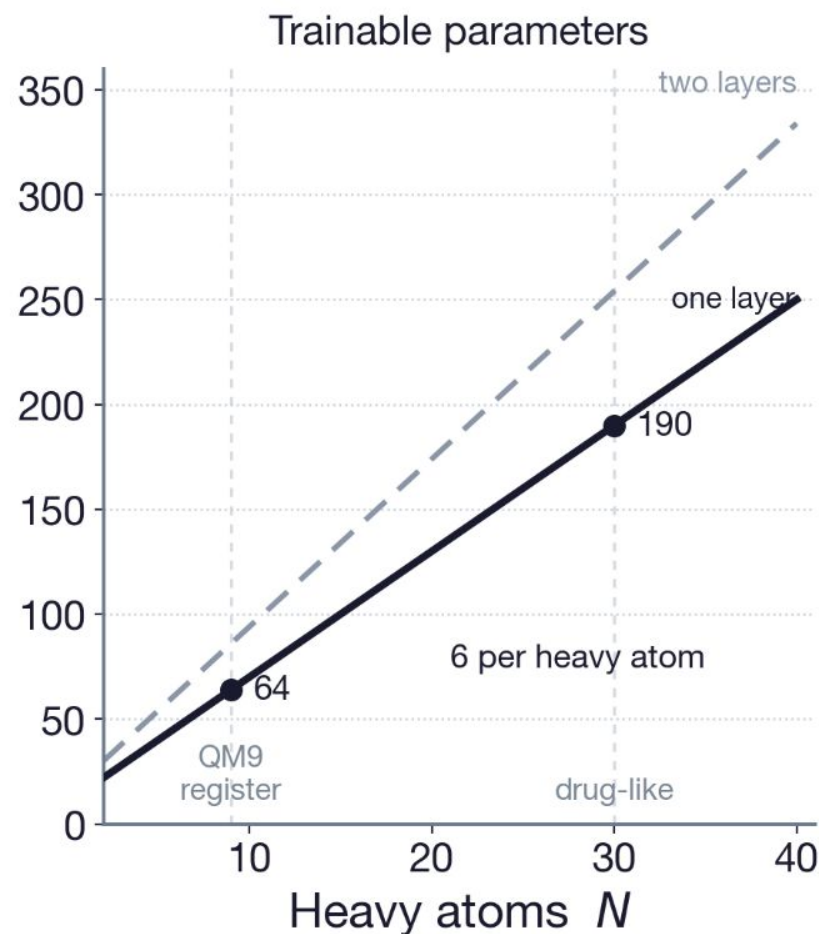
Conclusions & Future Work

A Comparison of Classical and Quantum Models.



Conclusions & Future Work

Scaling Beyond QM9.



Resource scaling only. At one layer information travels one bond, so larger molecules may need a second layer, the dashed lines. Whether accuracy holds at larger N is open.

Thank you
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QUNASYS

We are Quantum Native.



<https://qunasys.com/en/>

Founded

2018 Tokyo
Co-Founder & CEO:
Tennin Yan

Focus

Quantum computer
algorithm/software
development

Advisors

Prof. Keisuke Fujii
+ 3 more faculties from
Osaka University

Employees

58 (incl. 24 Ph.D.)
of which 36 are
researchers/engineers

R&D

17 patents/patents
pending, 45 papers
published in NISQ /
FTQC related areas

Region

Japan and Europe